Skills Summary

Rates of Change: Table, Graph, Equation

Use this reference while completing your worksheet or when studying.

Skill 1 — Rate of change from a table

Rule: pick two rows. Rate = (change in the output row) ÷ (change in the input row), taking both differences in the same direction.

Example: Find the rate of change for the table $x = 0, 3$ with $y = 4, 19$.
Step 1. Two rows are enough for a linear table, so use the pair given and put the output change on top.
Step 2. Output change $19 - 4 = 15$; input change $3 - 0 = 3$; the rate is $15/3$.
Each unit of x adds this much y. $\boldsymbol{m = 5}$

Try it: A table gives $x = 0, 4$ with $y = 50, 26$. Find the rate of change.

check: $\boldsymbol{m = -6}$

Skill 2 — Rate of change from an equation

Rule: in $y = mx + b$ the rate of change is m, the number multiplying the input. The constant b is the starting value and never changes the rate.

Example: Find the rate of change of $y = 6x + 20$.
Step 1. Nothing needs solving — the rate is what one extra unit of x adds, so compare the value at x with the value at $x + 1$.
Step 2. $\big(6(x+1) + 20\big) - (6x + 20) = 6$, which is exactly the number multiplying x.
The 20 cancels, as the rule promises. $\boldsymbol{m = 6}$

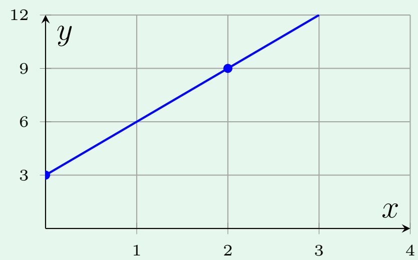
Try it: Find the rate of change of $y = 100 - 8x$.

check: $\boldsymbol{m = -8}$

Skill 3 — Rate of change from a graph

Rule: choose two points where the line crosses grid corners, then count the rise and the run between them. Rising left to right means a positive rate; falling means a negative one.

Example: Find the rate of change of the line drawn below.



Step 1. Two marked lattice points are enough, so read them off and count rise over run rather than guessing the steepness.
Step 2. From $(0, 3)$ to $(2, 9)$ the rise is $9 - 3 = 6$ and the run is $2 - 0 = 2$, giving $6/2$.
Rising left to right, so positive. $\boldsymbol{m = 3}$

Try it: A line passes through the marked points $(1, 2)$ and $(4, 8)$. Find its rate of change.

check: $\boldsymbol{m = 2}$

Skill 4 — Comparing two rates

Rule: convert both relationships to a rate in the *same units* first, then compare the two numbers. On a graph the larger rate is the steeper line.

Example: A table gives 5 dollars per hour; a job pays $y = 6x$ dollars for x hours. Which rate is greater?
Step 1. The two relationships arrive in different forms, so pull a rate out of each before comparing anything.
Step 2. Table rate 5 dollars per hour; equation rate is the number multiplying x, so 6 dollars per hour.
Table rate first, then equation rate. $\mathbf{5 < 6}$

Try it: A graph rises 9 metres per second; another runner follows $d = 4t$. Compare the graph's rate with the runner's.

check: $\mathbf{9 > 4}$